

TECHNOLOGY STACK

DATE	29 September
TEAM ID	SWTID-2026-1812
PROJECT NAME	Comic craft Ai comic story creater using gemini model
MAXIMUM MARK	2 Marks

1. Technology Stack

Architecture Component	Technology Chosen	Purpose
Front-End	HTML5	To create the structure and input interface of the web application.
Styling	CSS3	To provide a simple, attractive, and user-friendly interface.
Template Engine	Jinja2	To display dynamically generated comic content on web pages.
Back-End Framework	FastAPI	To handle application requests, processing, routing, and communication between components.
Programming Language	Python	To implement the application logic and integrate AI services.
AI Story Generation	Google Gemini API	To generate the comic story outline, narration, and dialogue from the user's prompt.

Image Generation	Stable Diffusion / Image Generation API	To generate visual images for the comic panels based on the story.
PDF Generation	FPDF	To convert the generated comic panels and content into a downloadable PDF.
Configuration	Python-dotenv	To securely load API keys and configuration values from environment variables.
Development Server	Uvicorn	To run and test the FastAPI application locally.
Version Control	GitHub	To store, manage, and share the project source code and documentation.

2. Technology Selection and Purpose

Technology Chosen	Purpose in ComicCraft
Python	Used as the main programming language because it provides easy integration with AI and web development libraries.
FastAPI	Used to build the backend and manage communication between the user interface, AI services, and output generation.
HTML & CSS	Used to design the comic creation form, preview page, and other web interfaces.
Jinja2	Used to dynamically display generated story details, dialogue, narration, and comic panels.
Google Gemini	Used to understand the user's story prompt and generate a structured comic story with narration and dialogue.
Stable Diffusion / Image Generation	Used to create visual content for the individual comic panels.

FPDF

Used to create the final downloadable comic PDF.

GitHub

Used for source-code management, project submission, and collaboration among team members.

3. Architecture Components

The ComicCraft system follows a simple web-based architecture where the user interacts with the front-end, while the FastAPI backend manages AI processing and output generation.

Main Components

1. User Interface

- Accepts story prompt, character name, setting, tone, and art style.
- Provides the interface for generating and viewing the comic.

2. FastAPI Backend

- Receives user requests.
- Coordinates story generation, image generation, preview creation, and PDF export.

3. Gemini AI Module

- Generates the five-panel story outline.
- Produces narration and dialogue for the comic.

4. Image Generation Module

- Creates images based on the descriptions of each comic panel.

5. Comic Preview Module

- Combines the generated text and images.
- Displays the completed comic to the user.

6. PDF Export Module

- Converts the generated comic into a downloadable PDF document.

7. Storage / Output Module

- Stores generated panel images and exported PDF files during application use.

4. Overall Architecture Flow

User → Web Interface → FastAPI Backend → Gemini AI → Image Generation → Comic Preview → PDF Export

This architecture allows the different components to work together as a single AI-powered comic creation system.